

Clinical safety dashboard — Impact of balancing on glucose prediction accuracy

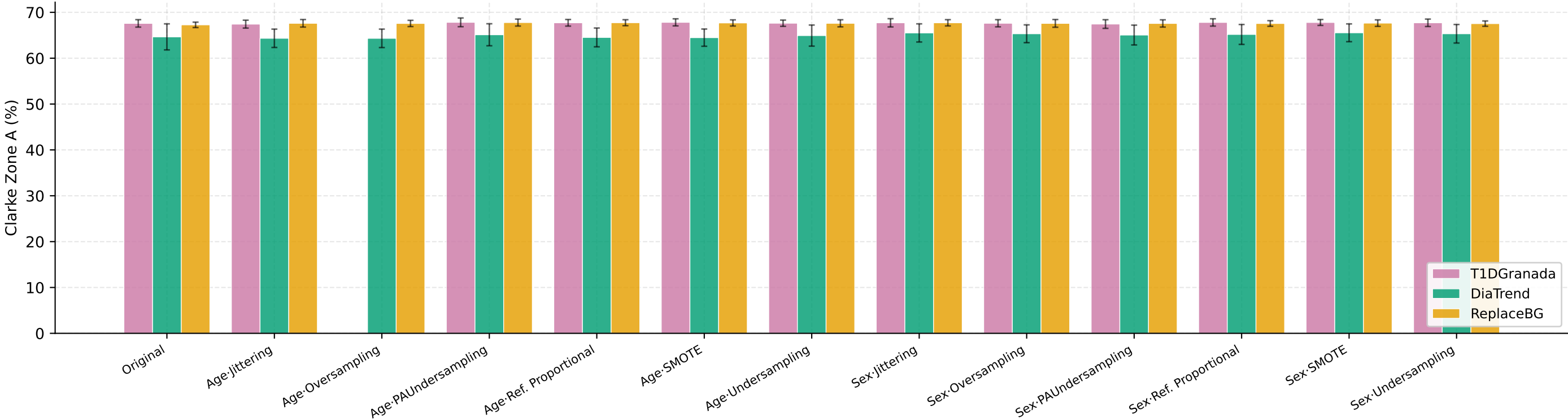
A | RMSE improvement · Hypo L2 (<54 mg/dL) — CRITICAL

	T1DGranada	DiaTrend	ReplaceBG
Age·Jittering	-1.7%	+3.2%	+1.7%
Age·Oversampling		+4.1%	+1.9%
Age·PAUndersampling	+0.3%	+2.2%	+0.9%
Age·Ref. Proportional	-0.8%	+3.3%	+0.9%
Age·SMOTE	+0.3%	+3.8%	+1.9%
Age·Undersampling	-1.1%	+0.8%	+0.2%
Sex·Jittering	+0.0%	+2.7%	+0.9%
Sex·Oversampling	+0.0%	+2.2%	+1.3%
Sex·PAUndersampling	-0.5%	+0.8%	+0.9%
Sex·Ref. Proportional	+0.1%	+1.7%	+0.5%
Sex·SMOTE	+0.0%	+2.9%	+1.1%
Sex·Undersampling	-0.9%	+1.7%	+0.3%

B | RMSE improvement · Hypo L1 (54-69 mg/dL)

	T1DGranada	DiaTrend	ReplaceBG
Age·Jittering	-0.6%	+3.1%	+2.4%
Age·Oversampling		+5.0%	+2.3%
Age·PAUndersampling	+0.7%	+2.0%	+1.4%
Age·Ref. Proportional	+0.2%	+4.5%	+0.9%
Age·SMOTE	+1.1%	+4.6%	+2.2%
Age·Undersampling	+0.5%	+0.6%	+0.3%
Sex·Jittering	+1.0%	+2.9%	+1.3%
Sex·Oversampling	+1.1%	+1.9%	+2.0%
Sex·PAUndersampling	+0.0%	+0.7%	+1.5%
Sex·Ref. Proportional	+1.3%	+1.5%	+0.3%
Sex·SMOTE	+1.0%	+2.3%	+1.3%
Sex·Undersampling	+0.3%	+1.4%	+0.2%

C | Clarke Zone A (%) — clinically safe predictions ↑ better



D | RMSE improvement · In Range (70-180 mg/dL)

	T1DGranada	DiaTrend	ReplaceBG
Age·Jittering	+0.4%	+4.4%	+0.1%
Age·Oversampling		+5.3%	+0.5%
Age·PAUndersampling	-0.7%	+1.1%	-0.5%
Age·Ref. Proportional	+0.0%	+4.8%	-1.1%
Age·SMOTE	-0.8%	+4.1%	+0.3%
Age·Undersampling	-0.3%	+0.8%	+0.1%
Sex·Jittering	+0.3%	+1.7%	+0.0%
Sex·Oversampling	+0.5%	+1.8%	-0.4%
Sex·PAUndersampling	+0.6%	+2.1%	-0.2%
Sex·Ref. Proportional	+0.2%	+2.0%	-0.5%
Sex·SMOTE	+0.0%	+1.9%	+0.2%
Sex·Undersampling	+0.4%	+1.8%	+0.1%

E | RMSE improvement · Hyper L1 (181-250 mg/dL)

	T1DGranada	DiaTrend	ReplaceBG
Age·Jittering	-1.2%	-4.7%	-1.8%
Age·Oversampling		-6.5%	-1.3%
Age·PAUndersampling	+0.1%	-0.6%	-0.3%
Age·Ref. Proportional	-0.3%	-5.0%	+1.2%
Age·SMOTE	+0.4%	-5.7%	-1.1%
Age·Undersampling	-0.9%	-1.0%	-0.4%
Sex·Jittering	-0.8%	-1.3%	-0.5%
Sex·Oversampling	-1.2%	-0.8%	-0.3%
Sex·PAUndersampling	-0.7%	-1.9%	-0.8%
Sex·Ref. Proportional	-0.4%	-1.5%	+0.1%
Sex·SMOTE	-0.5%	-0.9%	-0.9%
Sex·Undersampling	-0.1%	-0.8%	-0.2%